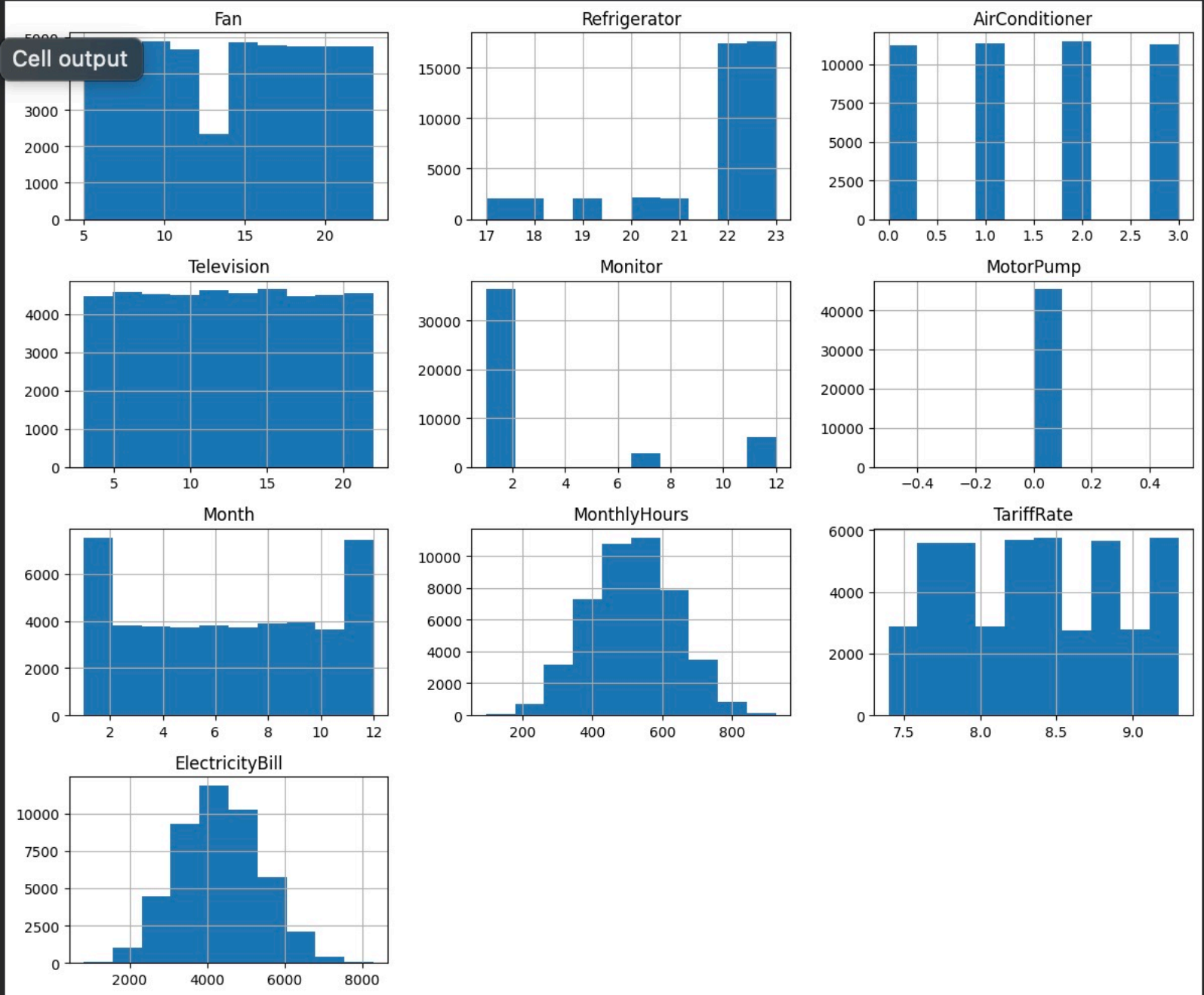
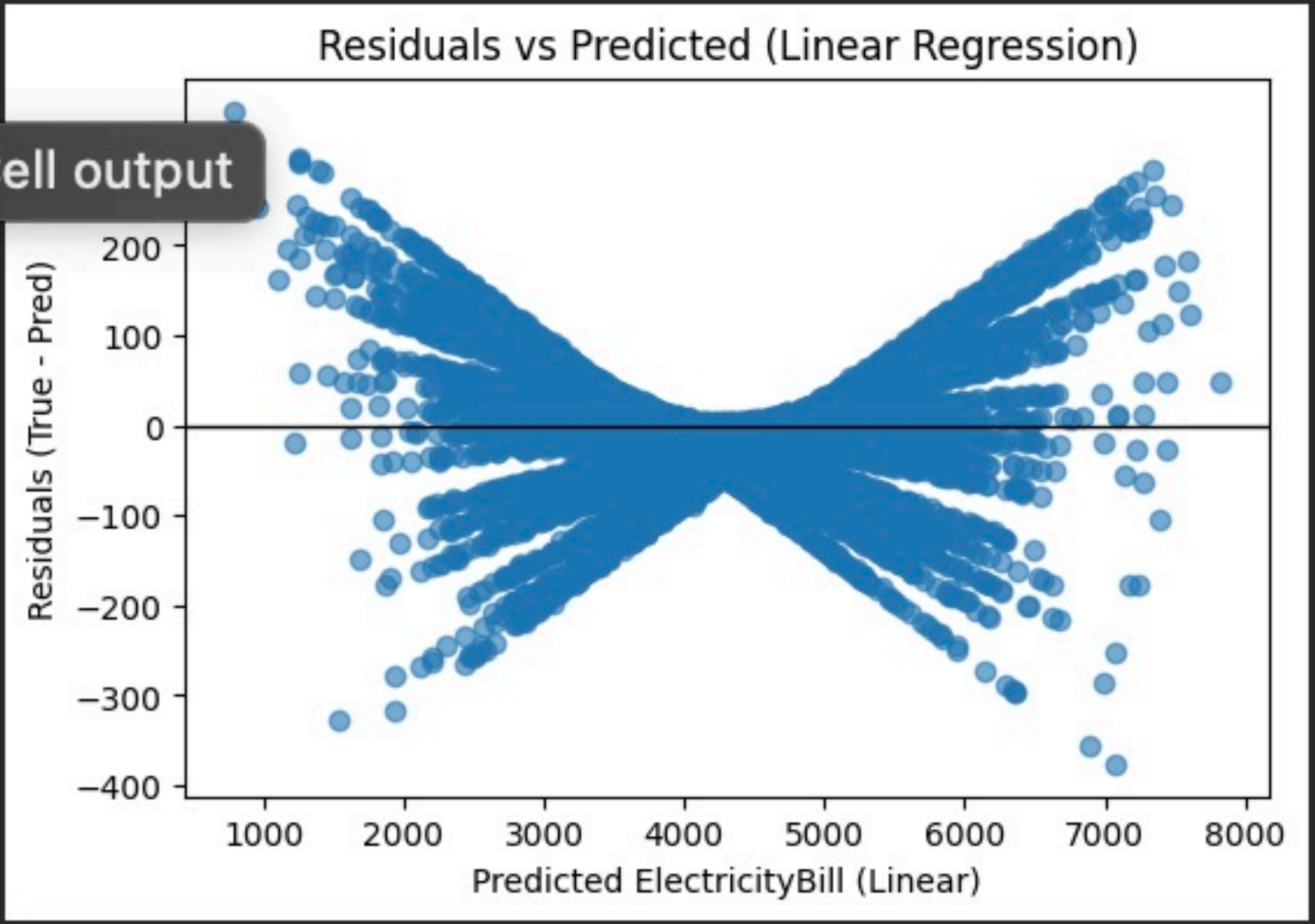
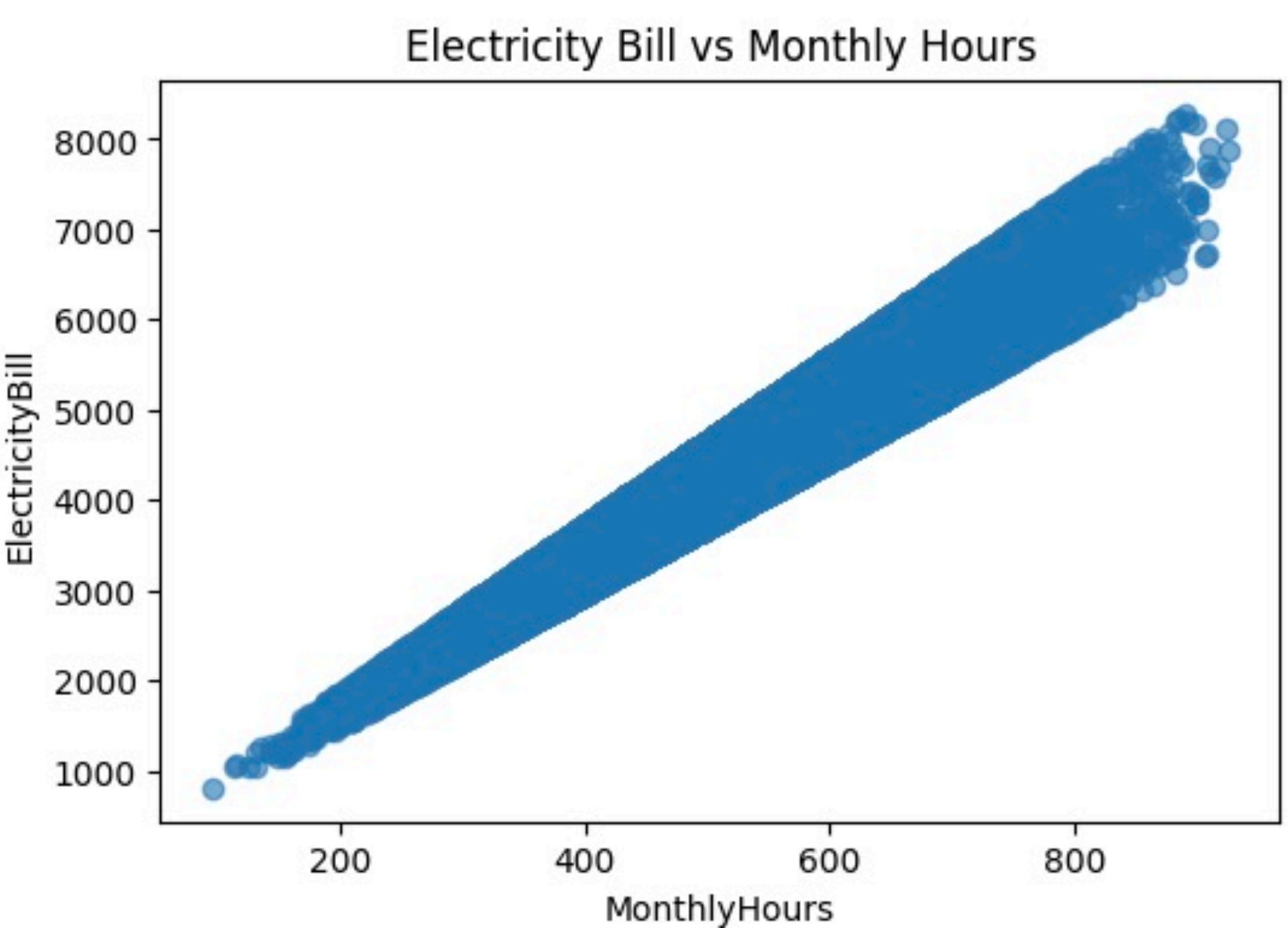
Cell output

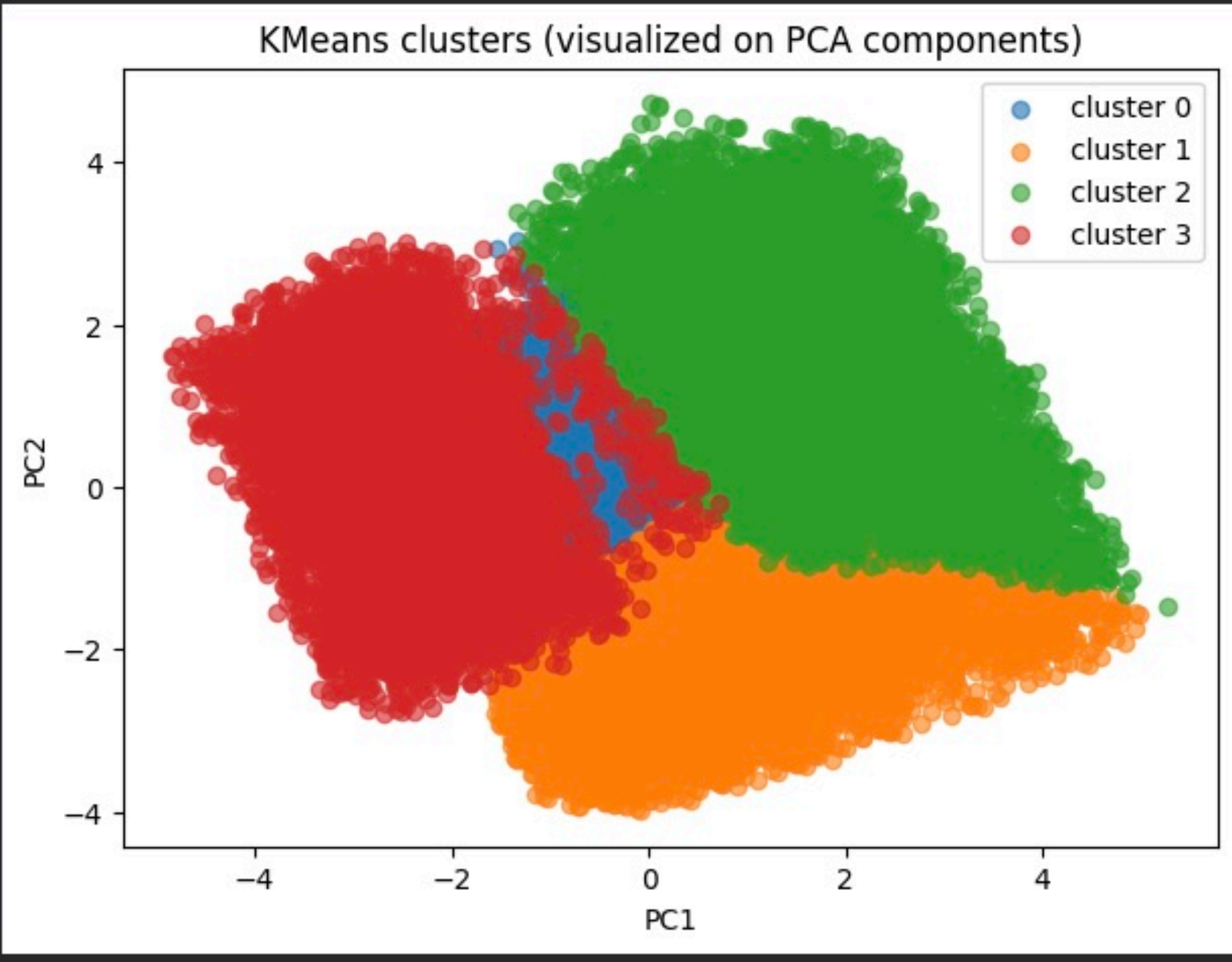

Fan
Refrigerator
AirConditioner
Television
Monitor
MotorPump
Month
MonthlyHours
TariffRate
ElectricityBill

Cell output




```
{'RMSE': np.float64(70.55840306977416),
 'MAE': 49.27244027860249,
 'R2': 0.9956306781863746}
```


2000
4000
6000
8000
Electricity Bill vs Monthly Hours
8000
7000
6000
5000
4000
3000
2000
1000
ElectricityBill
200
400
600
800
MonthlyHours


KMeans clusters (visualized on PCA components)
cluster 0
cluster 1
cluster 2
cluster 3
PC1
PC2
4
2
0
−2
−4
−4
−2
0
2
4

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Fan	45345.0	NaN	NaN	NaN	13.990694	5.470816	5.0	9.0	14.0	19.0	23.0
Refrigerator	45345.0	NaN	NaN	NaN	21.705458	1.672575	17.0	22.0	22.0	23.0	23.0
AirConditioner	45345.0	NaN	NaN	NaN	1.503959	1.115482	0.0	1.0	2.0	2.0	3.0
Television	45345.0	NaN	NaN	NaN	12.502635	5.756007	3.0	7.0	13.0	17.0	22.0
Monitor	45345.0	NaN	NaN	NaN	2.865057	3.894933	1.0	1.0	1.0	1.0	12.0
MotorPump	45345.0	NaN	NaN	NaN	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Month	45345.0	NaN	NaN	NaN	6.488058	3.443252	1.0	3.0	6.0	9.0	12.0
City	45345	16	Hyderabad	2914	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Company	45345	32	Ringfeder Power Transmission India Pvt. Ltd.	1486	NaN	NaN	NaN	NaN	NaN	NaN	NaN
MonthlyHours	45345.0	NaN	NaN	NaN	515.083207	122.618017	95.0	429.0	515.0	601.0	926.0
TariffRate	45345.0	NaN	NaN	NaN	8.369648	0.576992	7.4	7.9	8.4	8.9	9.3
ElectricityBill	45345.0	NaN	NaN	NaN	4311.771307	1073.886406	807.5	3556.8	4299.4	5038.8	8286.3

Cell output

Anomalies found: 454

l output

	Fan	Refrigerator	AirConditioner	Television	Monitor	MotorPump	Month	City	Company	MonthlyHours	TariffRate	ElectricityBill	TotalApplianceHours	Fan_share	Refrigerator_share	AirConditioner_share
136	7	18.0	0.0	21.0	7.0	0	12	New Delhi	Sterlite Power Transmission Ltd	476	8.5	4046.0	53.0	0.132075	0.339623	0.00000
279	9	17.0	3.0	5.0	7.0	0	9	New Delhi	NHPC	383	8.5	3255.5	41.0	0.219512	0.414634	0.07317
341	8	17.0	3.0	3.0	1.0	0	5	Chennai	NHPC	376	8.8	3308.8	32.0	0.250000	0.531250	0.09375
631	5	17.0	0.0	21.0	7.0	0	12	Faridabad	Orient Green	503	8.1	4074.3	50.0	0.100000	0.340000	0.00000
701	5	18.0	3.0	3.0	1.0	0	4	Shimla	Toshiba Transmission & Distribution Systems (I...	309	7.7	2379.3	30.0	0.166667	0.600000	0.10000